

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location AL facility AL_way_1540172608 -- station KTHA, America/Chicago
Building OSM 1540172608
OSM way 1540172608
Facade gap not applicable -- one building, no second facade
Weather record 42,489 real hourly records from KTHA
Plume physics NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2021-01-01 (recirc_edge)
Plant limit 21.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 8 of 24 hours with 1 mode change. That is 2 more than the reactive incumbent, at 0 unsafe hours against its 0. 5 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 8 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 6 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
5 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	11.000	21.0	11.000	0.702
01	mechanical	yes	switch budget	11.500	21.0	11.000	1.079
02	mechanical	yes	switch budget	13.500	21.0	15.000	1.012
03	mechanical	yes	switch budget	13.500	21.0	15.000	0.931
04	mechanical	yes	switch budget	13.500	21.0	15.000	0.888
05	mechanical	no	dew point	16.500	21.0	17.000	0.804
06	mechanical	no	dew point	17.000	21.0	17.000	0.968
07	mechanical	no	dew point	19.000	21.0	17.000	1.914

08	mechanical	no	dew point	22.000	21.0	18.000	2.378
09	mechanical	no	dew point	23.500	21.0	19.000	2.824
10	mechanical	no	dew point	23.000	21.0	20.000	2.136
11	mechanical	no	dew point	23.000	21.0	21.000	1.681
12	mechanical	no	dry-bulb	23.000	21.0	22.000	1.092
13	mechanical	no	dry-bulb	23.000	21.0	22.000	0.983
14	mechanical	no	dry-bulb	23.000	21.0	22.000	0.851
15	mechanical	no	dry-bulb	22.500	21.0	21.000	0.816
16	FREE	yes	-	21.000	21.0	19.000	0.892
17	FREE	yes	-	19.500	21.0	17.000	0.983
18	FREE	yes	-	18.500	21.0	16.000	1.606
19	FREE	yes	-	16.000	21.0	14.000	1.451
20	FREE	yes	-	14.000	21.0	11.000	1.935
21	FREE	yes	-	12.500	21.0	9.000	1.550
22	FREE	yes	-	11.500	21.0	9.000	1.139
23	FREE	yes	-	10.000	21.0	9.000	0.838

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.000 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.500 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.500 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.500 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.500 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.500 C would have passed. The outside air is too HUMID: the dew-point bound is 15.50 C against a 15.0 C maximum, failing by 0.50 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

06:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 17.000 C would have passed. The outside air is too HUMID: the dew-point bound is 16.00 C against a 15.0 C maximum, failing by 1.00 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

07:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 19.000 C would have passed. The outside air is too HUMID: the dew-point bound is 17.00 C against a 15.0 C maximum, failing by 2.00 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

08:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 22.000 C would have passed. The outside air is too HUMID: the dew-point bound is 18.50 C against a 15.0 C maximum, failing by 3.50 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

09:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 23.500 C would have passed. The outside air is too HUMID: the dew-point bound is 18.50 C against a 15.0 C maximum, failing by 3.50 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

10:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 23.000 C would have passed. The outside air is too HUMID: the dew-point bound is 17.50 C against a 15.0 C maximum, failing by 2.50 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

11:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 23.000 C would have passed. The outside air is too HUMID: the dew-point bound is 18.00 C against a 15.0 C maximum, failing by 3.00 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.000 C against a 21.0 C limit, so it fails by 2.000 C. It would take a limit 2.000 C higher, or a bound 2.000 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.000 C against a 21.0 C limit, so it fails by 2.000 C. It would take a limit 2.000 C higher, or a bound 2.000 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.000 C against a 21.0 C limit, so it fails by 2.000 C. It would take a limit 2.000 C higher, or a bound 2.000 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.500 C against a 21.0 C limit, so it fails by 1.500 C. It would take a limit 1.500 C higher, or a bound 1.500 C tighter, to change this hour.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 21.000 C, which is 0.000 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.892 C, and the margin is measured, not chosen: 0.892 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind

direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.500 C, which is 1.500 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.983 C, and the margin is measured, not chosen: 0.983 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.500 C, which is 2.500 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.606 C, and the margin is measured, not chosen: 1.606 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.000 C, which is 5.000 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.451 C, and the margin is measured, not chosen: 1.451 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.000 C, which is 7.000 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.935 C, and the margin is measured, not chosen: 1.935 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.500 C, which is 8.500 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.550 C, and the margin is measured, not chosen: 1.550 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.500 C, which is 9.500 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.139 C, and the margin is measured, not chosen: 1.139 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.000 C, which is 11.000 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.838 C, and the margin is measured, not chosen: 0.838 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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